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```
clear;
clc;
close all;
```

FILTER SPECIFICATIONS

```
Fs = 8000;              % Sampling frequency (Hz)
fc = 1500;              % Cutoff frequency (Hz)
wc = fc / (Fs/2);       % Normalized cutoff frequency
N = 50;                 % Filter order
```

WINDOW FUNCTIONS

```
w_rect  = rectwin(N+1);
w_hamm  = hamming(N+1);
w_hann  = hann(N+1);
w_black = blackman(N+1);
```

FIR FILTER DESIGN USING FIR1

```
b_rect  = fir1(N, wc, w_rect);
b_hamm  = fir1(N, wc, w_hamm);
b_hann  = fir1(N, wc, w_hann);
b_black = fir1(N, wc, w_black);
```

FREQUENCY RESPONSE CALCULATION

```
[H_rect, f] = freqz(b_rect, 1, 1024, Fs);
[H_hamm, ~] = freqz(b_hamm, 1, 1024, Fs);
[H_hann, ~] = freqz(b_hann, 1, 1024, Fs);
[H_black, ~] = freqz(b_black, 1, 1024, Fs);
```

MAGNITUDE RESPONSE IN dB

```
mag_rect  = 20*log10(max(abs(H_rect), 1e-10));
mag_hamm  = 20*log10(max(abs(H_hamm), 1e-10));
```

```
mag_hann  = 20*log10(max(abs(H_hann), 1e-10));
mag_black = 20*log10(max(abs(H_black), 1e-10));
```

DISPLAY NUMERICAL OUTPUTS

```
fprintf('=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===\n\n');

fprintf('Filter Order (N): %d\n', N);
fprintf('Sampling Frequency: %d Hz\n', Fs);
fprintf('Cutoff Frequency: %d Hz\n\n', fc);

fprintf('--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_rect(1:5));

fprintf('--- HAMMING WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_hamm(1:5));

fprintf('--- HANN WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_hann(1:5));

fprintf('--- BLACKMAN WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_black(1:5));
```

```
=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===

Filter Order (N): 50
Sampling Frequency: 8000 Hz
Cutoff Frequency: 1500 Hz

--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---
   -0.0118         0    0.0128    0.0103   -0.0058

--- HAMMING WINDOW COEFFICIENTS (First 5) ---
   -0.0009         0    0.0012    0.0011   -0.0008

--- HANN WINDOW COEFFICIENTS (First 5) ---
   1.0e-03 *

         0         0    0.2009    0.3592   -0.3588

--- BLACKMAN WINDOW COEFFICIENTS (First 5) ---
   1.0e-03 *

         0         0    0.0743    0.1374   -0.1434
```

GRAPH 1: MAGNITUDE RESPONSE COMPARISON

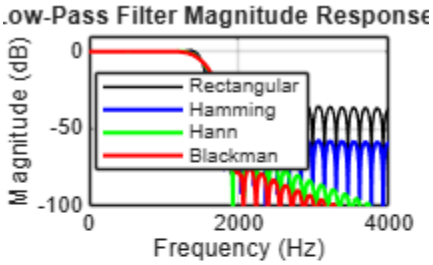
```
figure('Name', 'FIR Windowing Techniques Comparison');

subplot(2,1,1);
```

```
plot(f, mag_rect, 'k', 'LineWidth', 1);
hold on;
plot(f, mag_hamm, 'b', 'LineWidth', 1.5);
plot(f, mag_hann, 'g', 'LineWidth', 1.5);
plot(f, mag_black, 'r', 'LineWidth', 1.5);

grid on;
legend('Rectangular', 'Hamming', 'Hann', 'Blackman', ...
    'Location', 'best');

title('FIR Low-Pass Filter Magnitude Response (N = 50)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([0 Fs/2]);
ylim([-100 10]);
```

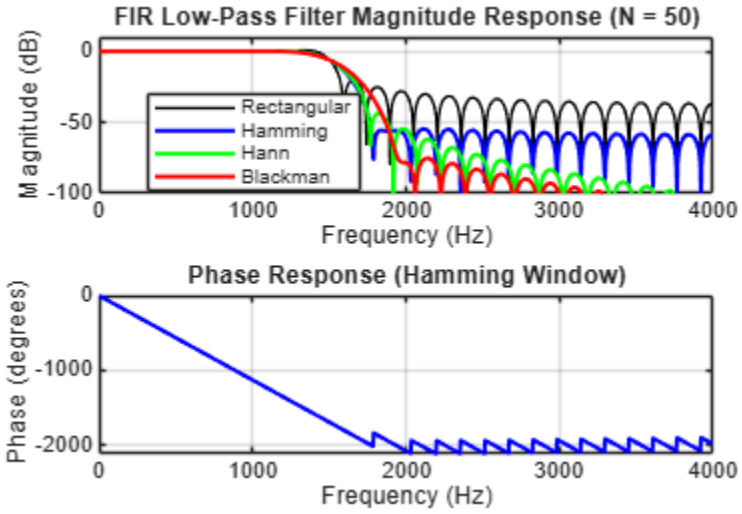


GRAPH 2: PHASE RESPONSE

```
subplot(2,1,2);

plot(f, unwrap(angle(H_hamm))*180/pi, ...
    'b', 'LineWidth', 1.5);

grid on;
title('Phase Response (Hamming Window)');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);
```



FINAL RESULT

```
fprintf('\n=============================================\n');
fprintf('FINAL RESULT\n');
fprintf('=============================================\n');

fprintf('1. Rectangular window FIR filter designed.\n');
fprintf('2. Hamming window FIR filter designed.\n');
fprintf('3. Hann window FIR filter designed.\n');
fprintf('4. Blackman window FIR filter designed.\n');
fprintf('5. Magnitude responses compared successfully.\n');
fprintf('6. Phase response plotted using Hamming window.\n');
```

```
=============================================
FINAL RESULT
=============================================
1. Rectangular window FIR filter designed.
2. Hamming window FIR filter designed.
3. Hann window FIR filter designed.
4. Blackman window FIR filter designed.
5. Magnitude responses compared successfully.
6. Phase response plotted using Hamming window.
```

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